Ayan Nath

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Education

Massachusetts Institute of Technology — Cambridge, MA
Ph.D. in Mathematics — September 2024 – present
Advisors: Bjorn Poonen, Keerthi Madapusi

Chennai Mathematical Institute — Chennai, India
Bachelor of Science (Honours) in Mathematics and Computer Science — September 2021 – April 2024

Research

- **Finite Generation in Cohomology of Artin Stacks** (in progress) 2026
- **Perfectoid Shimura Varieties of Exceptional Type** (coming very soon) 2026
  We show that all minimally compactified Shimura varieties at infinite level, including those of exceptional type, are perfectoid spaces for all sufficiently large primes.
- **Compactification of Reductive Group Schemes** [arXiv] 2026
  We prove that any isotrivial reductive group scheme over an arbitrary base admits a smooth projective equivariant compactification, verifying a conjecture of Česnavičius. In the adjoint case, we recover fiberwise the wonderful compactification. Finally, we give an example of a non-isotrivial torus admitting no equivariant compactification.
- **The Moments of Split Greatest Common Divisors** (with Abhishek Jha and Emanuele Tron)
  to appear in Acta Arithmetica [arXiv] 2024
- **On the Least Common Multiple of Polynomial Sequences at Prime Arguments** (with Abhishek Jha)
  International Journal of Number Theory, 18(06), [doi] 2022
- **On Quotients of Values of Euler's Function on Factorials** (with Abhishek Jha)
  Bulletin of the Australian Mathematical Society, 105(3), [doi] 2021
- **On the divisibility** $a! + b! \mid (a + b)!$
  The American Mathematical Monthly, 129(3), [doi] 2022

Invited Talks

- **Perfectoid Shimura Varieties of Exceptional Type** October 19, 2026
  Boston University Number Theory Seminar

Service

- **Research Science Institute (RSI)**, MIT Summer 2026
  Mentor for two high school students pursuing original research projects; one student won a top 5 best paper award amongst all disciplines being the only one from math.
- **Directed Reading Program**, MIT January 2026
  Mentor for three undergraduate students on complex elliptic curves and their moduli.

- **Directed Reading Program**, MIT — January 2025
  Mentor for two undergraduate students on the Kronecker-Weber theorem and Artin reciprocity.
- **Teaching Assistant**, Chennai Mathematical Institute
  - **Calculus 1** (Multivariable differential calculus) — January 2024 – April 2024
  - **Calculus 2** (Multivariable integral calculus) — August 2023 – December 2023
  - **Analysis 2** (Topology, function spaces, Fourier analysis) — August 2023 – December 2023
  - **Discrete Mathematics** — January 2023 – April 2023

AWARDS

2024 **Akamai Presidential Graduate Fellowship**, MIT.
CMI Medal of Excellence, top of class.

2022 **SRIRAM Scholarship**, tuition fee waiver and monthly stipend for undergraduate studies.
Spirit of Ramanujan
Indian National Mathematical Olympiad Awardee (2019, 2020, 2021), Homi Bhabha Centre For Science Education